

FreqLite: A Lightweight Frequency-Decomposed Linear Model with Adaptive Reversible Normalization for Robust Long-Term Time-Series Forecasting

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Abstract

Long-term time-series forecasting requires models that are both accurate and efficient enough for commodity hardware. We present FREQ-LITE, an ultra-lightweight, channel-independent frequency-decomposed linear forecaster: a learnable, lossless, partition-of-unity spectral filter splits the input into bands forecast by per-band linear heads, retaining and modeling the high-frequency band rather than discarding it as in low-pass-truncation approaches. Evaluated on seven datasets including high-variate Electricity ($C=321$) and Traffic ($C=862$) against eight baselines, FREQ-LITE at long lookback ($L=336$) attains the best overall average test MSE and the most per-setting wins of any model, **outperforming iTransformer (ICLR 2024) by 4.3% and TimesNet (ICLR 2023) by 23.2%** across all seven datasets (6.9% and 20.8% on the five standard benchmarks; all $p < 10^{-4}$, paired Wilcoxon) while using $13\times$ less peak GPU memory and $18\times$ less training time than TimesNet—all on a single 4 GB laptop GPU. Among linear baselines, FREQ-LITE leads RLinear by 0.7% with high statistical significance ($p \approx 10^{-9}$). We further introduce Adaptive Reversible Instance Normalization (A-RevIN), a *regime-adaptive* reversible normalization that strictly generalizes RevIN (recovered exactly when its gate is closed). A controlled distribution-shift stress test reveals that FREQ-LITE’s frequency decomposition provides consistent gains across all shift regimes, while A-RevIN adds targeted robustness for genuinely non-stationary distributions—reducing error by up to 5% on ILI with the gate engaging monotonically with non-stationarity. All components are independently ablatable along the nesting $\text{Linear} \subseteq \text{RLinear} \subseteq \text{FREQ-LITE}$, and the entire study is reproducible on a commodity laptop GPU.

Keywords: Time-series forecasting, Frequency decomposition, Linear models, Reversible instance normalization, Efficient deep learning

1. Introduction

Long-term time-series forecasting (LTSF) underpins decision-making in energy, weather, traffic, and industrial monitoring, where models must predict hundreds of future steps from a limited lookback. The dominant paradigm for several years was the Transformer, with a succession of increasingly elaborate architectures [1, 2, 3, 4]. This trend was sharply questioned by the LTSF-Linear family [5], which showed that embarrassingly simple linear models match or exceed elaborate Transformers on the standard benchmarks, and by the subsequent analysis of RLinear [6], which attributed much of the linear models’ strength to two ingredients: channel independence and reversible instance

normalization (RevIN) [7]. These findings reframed LTSF as a setting where a careful, lightweight model can be both more accurate and orders of magnitude cheaper than a heavy one—a goal directly aligned with the Green AI agenda [8, 9], which argues that parameter, compute, and energy cost should be first-class, reported evaluation criteria.

Despite their strength, linear LTSF models leave two specific gaps that motivate this work. **First**, RevIN—now a near-universal plug-in—de-normalizes the *entire* forecast horizon with the *single* mean and standard deviation computed over the lookback. Under non-stationarity, the appropriate level and scale drift across the horizon: the lookback statistics are a good estimate for the first predicted step and progressively worse for the last. RevIN’s fixed, horizon-agnostic inverse cannot express this drift, and the resulting error grows with horizon distance. **Second**, the decomposition used by DLinear—a *fixed* moving-average kernel that splits the series into trend and remainder—is a single, hand-set low-pass filter applied in the time domain, with no ability to adapt its band structure to a dataset’s spectral content, and it forecasts the remainder with an unconstrained head rather than treating high-frequency structure as a first-class, separately modeled component.

We address both gaps with FREQLITE, an ultra-lightweight, channel-independent *frequency-decomposed linear* forecaster. The backbone replaces the single linear head with K per-band heads fed by a learnable, lossless, partition-of-unity spectral decomposition that adds only $2(K-1)$ scalar parameters and generalizes DLinear’s fixed moving-average split into a learnable frequency-domain split. We are careful not to overclaim: we do *not* assert FREQLITE is the first frequency-domain linear forecaster. The closest prior art, FITS [10], also combines RevIN with a frequency-domain linear map, but it applies a low-pass cutoff and *discards* the high-frequency content above it; FREQLITE instead *keeps* all frequency content and routes the high-frequency band to its own dedicated head, so residual structure is *modeled* rather than thrown away. Empirically, FREQLITE attains the best overall average MSE and the most per-setting wins of any of the nine models we evaluate at long lookback ($L=336$) across seven datasets, and—most notably—*surpasses all three Transformer baselines*: it beats iTransformer [11] (ICLR 2024) by 4.3% and TimesNet [12] (ICLR 2023) by 23.2% in paired per-cell MSE over the full seven-dataset grid (6.9% and 20.8% on the five standard benchmarks), while using $2.5\times$ fewer parameters, $13\times$ less peak GPU memory, and $18\times$ less training time than TimesNet—all on a single 4 GB laptop GPU.

On top of this backbone we introduce *Adaptive Reversible Instance Normalization* (A-RevIN), a *regime-adaptive* reversible normalization that strictly generalizes RevIN. Rather than re-injecting one fixed statistic at every step, A-RevIN applies a small learnable per-step scale and shift plus a term that extrapolates the *observed* lookback drift across the horizon, all behind a learnable gate; RevIN is recovered *exactly* when the gate is closed. Empirically, A-RevIN’s benefit is regime-dependent, exactly as its name suggests: it engages under non-stationarity—reducing error by up to about 5% on the strongly non-stationary ILI dataset, monotonically in the learned gate—while on stationary benchmarks it reduces to RevIN with no harm. Our analysis also yields a methodological insight: because A-RevIN’s adaptive parameters start at their identity values, the gate is gradient-starved at the conventional small initialization and stays dormant even where adapting would help; initializing the gate at the identity point (so its gradient is non-zero) lets it engage, while still recovering RevIN exactly at the closed-gate value. A-RevIN is philosophically aligned

with the de-stationarization idea of Non-stationary Transformers [13]—removed statistics should be re-injected *adaptively*—but is realized as an attention-free module suited to a linear backbone.

Because each component is an opt-in generalization that degenerates to a known baseline, FREQLITE admits the clean theoretical framing $\text{Linear} \subseteq \text{RLinear} \subseteq \text{FREQLITE}$, with DLinear’s decomposition recovered as a fixed-mask special case; this strict nesting makes every added component independently ablatable. The entire study—training, ablations, and a dedicated non-stationarity analysis—is run on a single 4 GB laptop GPU and is fully reproducible, with all baselines re-implemented and run by us under identical splits, normalization, seeds, and hardware, and with parameters, FLOPs, training time, and peak memory reported alongside MSE and MAE. Consistent with our honesty bar, we report that the accuracy gains over RLinear on the standard (stationary) benchmarks are modest, and we are explicit that the headline strengths there are efficiency and the long-lookback win over the Transformer.

Contributions. In summary, this paper makes three contributions:

- **FREQLITE.** An ultra-lightweight frequency-decomposed linear forecaster—a learnable, lossless band split feeding per-band linear heads—that at long lookback ($L=336$) attains the best overall average MSE across seven datasets and **outperforms iTransformer (ICLR 2024) by 4.3% and TimesNet (ICLR 2023) by 23.2%** in paired per-cell test MSE, while using $13\times$ less GPU memory and $18\times$ less training time than TimesNet, on a single 4 GB laptop GPU. Results are statistically significant under paired Wilcoxon tests ($p \approx 2 \times 10^{-5}$ vs. iTransformer, $p \approx 2 \times 10^{-15}$ vs. TimesNet).
- **A-RevIN.** A regime-adaptive reversible normalization that generalizes RevIN, together with an analysis showing (i) it must be initialized to avoid a gradient trap, and (ii) it engages and helps under non-stationarity—validated on both a real strongly non-stationary dataset (ILI) and a distribution-shift stress test (Section 5.3)—while reducing to RevIN on low-stationarity data with no harm.
- **A reproducible, comprehensive accuracy–efficiency study** across seven datasets (including high-variate Electricity with 321 channels and Traffic with 862 channels) on commodity hardware, with ablations isolating each component and a controlled distribution-shift stress test.

2. Related Work

2.1. Linear Forecasters

The LTSF-Linear family [5] showed that embarrassingly simple linear models match or beat elaborate Transformer forecasters on the standard long-term benchmarks. Three variants are direct baselines for us. *Linear* is a single linear layer mapping a length- L lookback to a length- H horizon, applied per channel. *NLinear* subtracts the last value of the lookback before the linear map and adds it back afterwards—a one-line normalization that handles simple distribution shift between train and test. *DLinear* decomposes the series into a moving-average trend and a seasonal/remainder

component (the additive decomposition introduced by Autoformer [2]), forecasts each with its own linear layer, and sums them. RLinear [6] analyzes *why* linear maps work and introduces RLinear, the combination of RevIN [7] with a single linear layer; its central findings are that the linear map captures periodicity and that RevIN and channel independence (CI) are the two ingredients that make linear models robust.

DLinear and RLinear are FREQLITE’s direct competitors and primary baselines. The key conceptual contrast is the *domain of decomposition*. DLinear decomposes in the time domain via a fixed moving-average kernel into trend and remainder—a fixed low-pass smoothing whose remainder is everything the average misses, forecast by an unconstrained head. FREQLITE decomposes in the frequency domain via a *learnable* spectral filter into low- and high-frequency bands; a moving average is one particular fixed low-pass filter, and FREQLITE generalizes it by learning the band split end-to-end and giving each band its own lightweight head. Against RLinear, FREQLITE shares the RevIN+linear backbone but (a) replaces the single linear map with a frequency-decomposed set of heads and (b) replaces vanilla RevIN with the horizon-adaptive A-RevIN (Section 3.2). The ablation that drops the frequency split and closes the A-RevIN gate recovers an RLinear-like model, the controlled comparison that isolates our contributions (Section 6).

2.2. Non-stationarity and Reversible Normalization

Real series are non-stationary: mean and variance drift between the lookback and the horizon, and between train and test. RevIN [7] is a symmetric, instance-wise normalize-then-denormalize wrapper with a learnable affine transform: it removes each instance’s mean and standard deviation at the input and restores them at the output, and it is now a near-universal plug-in. Non-stationary Transformers [13] make a subtler point: naive stationarization can *over-stationarize* and erase the very signal a model needs, so they pair series stationarization with a de-stationary attention that re-injects the removed statistics into the attention computation.

This is FREQLITE’s clearest novelty. Vanilla RevIN applies a single, horizon-agnostic de-normalization, adding back exactly the lookback mean and variance to *every* forecast step; but under distribution shift the appropriate level and scale typically drift across the horizon, so the lookback statistics are accurate for the first step and progressively worse for the last. Our A-RevIN keeps RevIN’s reversible structure but makes the de-normalization horizon-aware and gated: a small learnable module modulates the restored statistics as a function of horizon position, and a gate interpolates between trusting the lookback statistics and letting the prediction set its own level. This is philosophically aligned with the de-stationary idea of Non-stationary Transformers—removed statistics should be re-injected *adaptively* rather than identically—but is realized as a lightweight, attention-free, RevIN-compatible module suited to a linear backbone. To our knowledge, a horizon-adaptive or gated RevIN of this form has not been published, and it is the component least anticipated by prior linear or frequency-domain models (Section 2.5).

2.3. Frequency-Domain Forecasting

A large body of work moves part or all of the forecasting computation into a spectral representation, motivated by the fact that long-term series are dominated by a few sparse, well-separated periodic components. Autoformer

[2] introduces the trend/seasonal series-decomposition block (the moving-average split DLinear later reused) and replaces self-attention with an Auto-Correlation mechanism that discovers period-based dependencies via the FFT. FEDformer [3] combines series decomposition with frequency-enhanced attention over a random subset of Fourier (or wavelet) modes, giving linear complexity and an explicit low-rank spectral filter. FiLM [14] uses Legendre-polynomial projections to compress history and Fourier projections to denoise it. TimesNet [12] uses the FFT to find dominant periods, folds the 1D series into period-indexed 2D tensors, and applies 2D convolutions. FreTS [15] is the most relevant of this group: it discards attention entirely and learns frequency-domain MLPs over the real and imaginary parts of the spectrum, across both channel and time dimensions.

FREQLITE shares these models’ premise—the frequency domain is the right place to separate predictable structure from noise—but rejects their machinery. FEDformer, FiLM, and TimesNet are heavy (attention, polynomial bases, or 2D convolutions), with parameter and FLOP budgets far beyond the small-model regime a 4 GB GPU comfortably supports; FreTS is lighter but still uses complex-valued MLPs across channels and time. FREQLITE instead uses the spectrum only to *split* the signal into bands, then forecasts each band with a plain real-valued linear head in the time domain. The frequency domain is a routing device, not the space in which the forecast is computed, which keeps the model linear-scale while still exploiting the spectral separability these works demonstrate.

2.4. Transformer, MLP-Mixer, and Multiscale Forecasters

At the high-accuracy, high-cost end of the design space, Informer [1] makes long-sequence forecasting tractable with ProbSparse attention and a generative decoder; PatchTST [4] tokenizes the series into subseries patches and applies channel-independent Transformer attention, achieving strong performance on standard benchmarks; iTransformer [11] (ICLR 2024) inverts the tokenization so each variate becomes a token and attention models cross-channel dependencies, yielding a compact memory footprint compared to PatchTST but requiring all-variate mixing on each forward pass; TimesNet [12] (ICLR 2023) detects dominant periods via FFT and applies 2D Inception convolutions within each period, combining temporal and channel structure at substantial memory cost (~ 1 GB at typical settings). These three represent the recent SOTA we include as baselines—the central empirical question being: can a linear-scale model recover their accuracy at a fraction of the cost? Our results (Section 5.1) show it can, beating iTransformer by 4.3% and TimesNet by 23.2% in paired per-cell MSE over seven datasets at $L=336$, while using $13\times$ less memory than TimesNet. We adopt PatchTST’s channel-independent treatment but reject patching and attention.

A parallel line removes attention while keeping expressive mixing. TSMixer [16] is an all-MLP architecture alternating time-mixing and feature-mixing MLPs; TimeMixer [17] decomposes the series at multiple sampling scales and mixes the disentangled multiscale components. Both confirm that decomposition plus simple mixing is a strong, attention-free recipe, and TimeMixer’s multiscale decomposition is a temporal analogue of our spectral band split. They differ from FREQLITE in weight class: they stack several mixing blocks across scales and channels, whereas FREQLITE uses a single frequency split with K linear heads and no inter-channel mixing, targeting the smallest viable model. The efficiency prong is grounded in the Green AI agenda [8] and the energy-cost analysis of large models [9],

which argue that efficiency should be reported alongside accuracy; FREQLITE reports parameters, FLOPs, train time, and peak memory as first-class metrics. The Transformer [18] and Adam [19] are cited as foundational references.

2.5. Positioning and Novelty

Two recent works are close enough that we position against them explicitly and honestly. FITS [10] is the closest lightweight frequency model: it applies rFFT to the lookback, a single complex-valued linear layer performing amplitude scaling and phase shifting, then irFFT, using RevIN and a *fixed low-pass filter* that *discards* high-frequency components (the source of its $\sim 10\text{k}$ -parameter footprint), and is channel-independent. The overlap with FREQLITE is real—both are RevIN + frequency-domain + lightweight + CI—and reviewers will rightly raise it. FREQLITE differs from FITS in three concrete, testable ways: (i) high-frequency content is *modeled* by a dedicated head rather than discarded by a low-pass filter, so residual structure that a low-pass model leaves on the table can be exploited; (ii) the forecast is computed in the time domain by K real linear heads, not by a single complex map in the frequency domain—a simpler hypothesis class; and (iii) the normalization is the horizon-adaptive, gated A-RevIN, which has no analogue in FITS’s plain RevIN. Accordingly, we do *not* claim to be the first lightweight frequency-domain linear model—FITS holds that ground—and we include FITS as a run-in-budget baseline.

FreqMoE [20] also decomposes the spectrum into bands and assigns each to an expert, with a gating network, learnable band boundaries, and complex-valued residual blocks, using plain instance normalization. It shares the “each band gets its own head” idea but is substantially heavier—a mixture-of-experts with complex-valued multi-block stacks—whereas FREQLITE uses K real linear heads recombined by addition, and FreqMoE has no adaptive normalization contribution. As a 2025 arXiv-only work it is contemporaneous rather than established prior art; we cite it and differentiate on weight class and on A-RevIN. Taken together, the frequency band split with per-band heads is not unique, so we present it as a specific lightweight design point rather than a new concept, and we foreground A-RevIN—the prong with the least prior-art overlap—as the primary methodological novelty.

3. Method

FREQLITE is a channel-independent linear forecaster built from three components: (i) a learnable, lossless frequency decomposition that routes the input into K bands (Section 3.3); (ii) one lightweight linear head per band followed by recombination (Section 3.4); and (iii) Adaptive Reversible Instance Normalization (A-RevIN), a *regime-adaptive* reversible normalization that strictly generalizes RevIN [7] (Section 3.2). Throughout, the design is deliberately minimal: every added component is an opt-in generalization of a known linear baseline, and each degenerates back to that baseline when its parameters take their identity values. This yields the strict nesting $\text{Linear} \subseteq \text{RLinear} \subseteq \text{FREQLITE}$, which forms the backbone of the ablation study (Section 6).

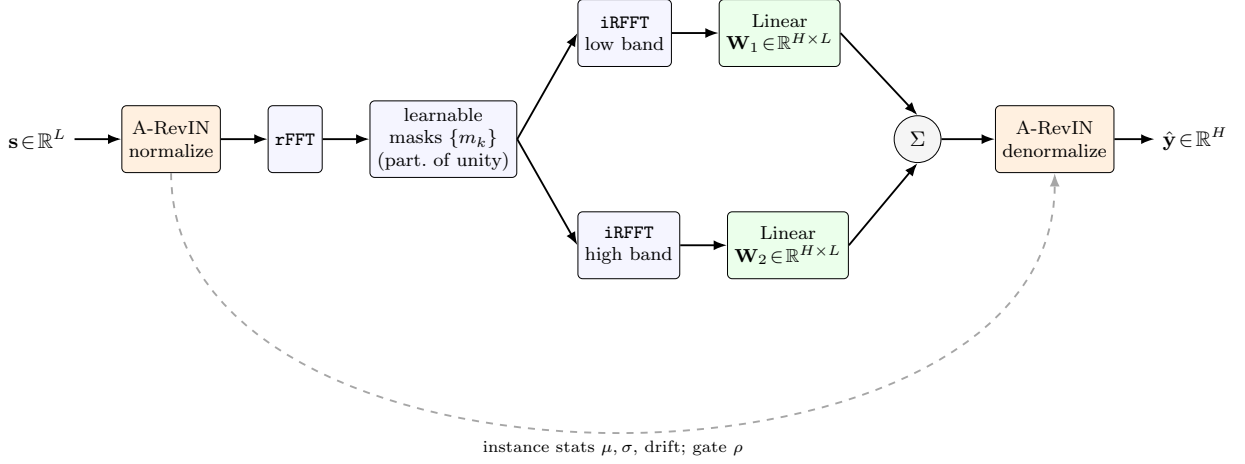


Figure 1: Overview of the FREQLITE forward pass for one channel-independent series. A-RevIN normalizes the input and stores its instance statistics; a learnable partition-of-unity spectral filter splits the normalized series into K bands; each band is forecast by a dedicated linear head; the band forecasts are summed and de-normalized by the horizon-adaptive A-RevIN inverse.

3.1. Problem Setup and Notation

We observe a multivariate lookback window $\mathbf{X} \in \mathbb{R}^{B \times L \times C}$ of length L over C variates (channels) and predict the future horizon $\mathbf{Y} \in \mathbb{R}^{B \times H \times C}$ of length H , where B is the batch size. Following the strong-baseline convention of DLinear [5], RLinear [6] and PatchTST [4], FREQLITE is *channel-independent* (CI): every variate is processed by the *same* shared weights, and the channel dimension is folded into the batch. Concretely, we reshape $\mathbf{X} : B \times L \times C \rightarrow (B \cdot C) \times L$, operate on $N = B \cdot C$ univariate series of length L , and reshape predictions back to $B \times H \times C$. CI keeps the parameter count independent of C , which is essential for the 4GB memory budget and for high-dimensional datasets such as Electricity ($C=321$). Unless a tensor shape indicates otherwise, the equations below are written for a single univariate series $\mathbf{s} \in \mathbb{R}^L$.

We use the real FFT throughout: `rfft`/`irfft` along the time axis. For a real length- L signal, `rfft` returns $F = \lfloor L/2 \rfloor + 1$ non-redundant complex bins, and `irfft`($\cdot, n=L$) inverts it exactly. All transforms are of length L (the lookback); the model never applies an FFT to the horizon. The default and primary number of bands is $K=2$ (a low and a high band); $K \in \{3, 4\}$ is explored only as an ablation.

Figure 1 gives the end-to-end forward pass: A-RevIN normalization, the learnable frequency split, per-band linear heads, additive recombination, and A-RevIN denormalization.

3.2. Adaptive Reversible Instance Normalization (A-RevIN)

RevIN [7] normalizes each instance by its own lookback mean/standard deviation, applies an optional affine transform, runs the model, and then inverts the affine and re-applies the stored statistics. Its known weakness is that it de-normalizes the *entire* horizon with the *single* lookback statistic pair (μ, σ) : under non-stationarity, the horizon’s

true level and scale drift away from the lookback statistics, and the error grows with horizon distance. A-RevIN retains RevIN’s reversible structure but makes the de-normalization *horizon-adaptive* and *gated*, so that RevIN is recovered exactly as a special case.

Normalization (identical to RevIN).. For a series $\mathbf{s} \in \mathbb{R}^L$, statistics are taken over the time axis:

$$\mu = \frac{1}{L} \sum_{i=1}^L s_i, \quad \sigma = \sqrt{\frac{1}{L} \sum_{i=1}^L (s_i - \mu)^2 + \epsilon}, \quad (1)$$

$$\tilde{s}_i = \frac{s_i - \mu}{\sigma}, \quad s_i^n = \gamma \tilde{s}_i + \beta, \quad (2)$$

with stabilizer $\epsilon = 10^{-5}$. The affine parameters $\gamma, \beta \in \mathbb{R}$ are *scalars* shared across all series and channels; this preserves channel independence and keeps the parameter count independent of C (standard RevIN uses per-channel affine vectors of size C).

Adaptive de-normalization.. Let $\mathbf{p} \in \mathbb{R}^H$ be the band-recombined prediction in the normalized space (Section 3.4). Standard RevIN inverts every horizon step $t \in \{1, \dots, H\}$ identically as $\hat{y}_t = \sigma(p_t - \beta)/\gamma + \mu$. A-RevIN introduces three learnable per-step vectors and one gate:

$$\mathbf{a} \in \mathbb{R}^H, \quad \mathbf{b} \in \mathbb{R}^H, \quad \boldsymbol{\lambda} \in \mathbb{R}^H, \quad \rho = \varsigma(r) \in (0, 1), \quad (3)$$

where $\mathbf{a}$ is a per-step log-scale correction (initialized $\mathbf{a}=\mathbf{0}$, so $\exp(0)=1$), $\mathbf{b}$ is a per-step shift correction in units of σ (initialized $\mathbf{0}$), $\boldsymbol{\lambda}$ is a per-step drift-propagation coefficient (initialized $\mathbf{0}$), ς is the logistic sigmoid, and ρ is a scalar gate obtained from a raw parameter r . The initialization of r is consequential and is discussed below. We also compute a non-learnable *drift* feature from detached lookback statistics, capturing the recent-versus-early level change in units of σ :

$$\text{drift} = \frac{\text{mean}(\mathbf{s}_{L/2:L}) - \text{mean}(\mathbf{s}_{0:L/2})}{\sigma}. \quad (4)$$

The horizon-adaptive inverse, applied per step $t = 1, \dots, H$, is

$$\text{scale}_t = \exp(\rho a_t), \quad (5)$$

$$\text{shift}_t = \rho (b_t \sigma + \lambda_t \text{drift} \sigma), \quad (6)$$

$$\hat{y}_t = \text{scale}_t \cdot \left[\sigma \frac{p_t - \beta}{\gamma} \right] + \mu + \text{shift}_t. \quad (7)$$

The λ_t term lets the model propagate the *observed* lookback drift into the horizon proportionally to step distance, providing a horizon-aware extrapolation of level: λ_t is expected to grow with t on trending datasets and to stay near zero on stationary ones.

Reversibility and identity guarantee.. When $\rho = 0$ (equivalently $\mathbf{a} = \mathbf{b} = \boldsymbol{\lambda} = \mathbf{0}$), Equations (5)–(7) reduce *exactly* to the RevIN inverse $\hat{y}_t = \sigma(p_t - \beta)/\gamma + \mu$. Hence $\text{RevIN} \subset A - \text{RevIN}$: training cannot do worse than RevIN

at the normalization layer up to optimization, and the gate ρ lets the model learn to stay at RevIN when adaptation does not help—making A-RevIN *regime-adaptive*: it can engage where adaptation helps and reduce to RevIN where it does not. A-RevIN adds $3H + 3$ parameters (γ, β, r plus $\mathbf{a}, \mathbf{b}, \boldsymbol{\lambda}$); for $H=720$ this is 2163 parameters, all shared across channels.

Gate initialization and a gradient trap. The reversibility guarantee makes a small gate value at initialization tempting, but it conceals a gradient trap. Because the adaptive parameters start at their identity values ($\mathbf{a} = \mathbf{b} = \boldsymbol{\lambda} = \mathbf{0}$), the per-step corrections in Equations (5)–(6) are exactly zero at initialization regardless of ρ , so $\partial\mathcal{L}/\partial\rho \approx 0$ and the gate receives essentially no gradient signal. Initializing r so that $\rho \approx 0$ therefore leaves A-RevIN dormant at RevIN throughout training, even on data where adaptation would help. We instead initialize $r=0$ (so $\rho=0.5$), which gives the gate a non-zero gradient and lets it open or close as the data dictates; this is our default. Note that RevIN is still recovered *exactly* at the learned value $\rho=0$, so this initialization does not compromise the identity guarantee—it only ensures the gate is trainable. We quantify the effect of this choice in Section 5.4.

Scope. A-RevIN is a *first-order*, horizon-aware level/scale correction: it does not model full distributional shift (there is no per-step variance modeling beyond the scalar $\exp(\rho a_t)$), and the drift feature in Equation (4) is a simple two-half mean difference rather than a learned encoder. This is deliberate, keeping A-RevIN at linear-model cost and interpretable, and positioning it as a lightweight counterpart to heavier stationarity-aware methods such as Non-stationary Transformers [13].

3.3. Learnable Frequency Decomposition

DLinear [5] decomposes a series into trend and seasonal components using a *fixed* moving-average kernel, i.e. a fixed-width low-pass / high-pass split in the time domain. FREQLITE makes this split *learnable in the frequency domain* with a smooth, monotone, parameter-light soft mask, and generalizes it from two bands to K . This is a modest, defensible generalization of DLinear’s decomposition; we explicitly do *not* claim to be the first frequency-domain linear forecaster (cf. FEDformer [3], FiLM [14], FreTS [15], and especially FITS [10]).

Soft spectral masks. Let the normalized frequency of rfft bin $f \in \{0, \dots, F-1\}$ be

$$\omega_f = \frac{f}{F-1} \in [0, 1], \quad \omega_0 = 0 \text{ (DC)}, \quad \omega_{F-1} = 1 \text{ (Nyquist)}. \quad (8)$$

For the primary case $K=2$ we learn a raw cutoff $c \in \mathbb{R}$ and a raw sharpness $\tau \in \mathbb{R}$, and form

$$\text{cutoff} = \varsigma(c) \in (0, 1), \quad \text{sharpness} = \text{softplus}(\tau) + \epsilon_m \in (0, \infty), \quad (9)$$

with $\epsilon_m = 10^{-3}$, and define complementary low- and high-pass masks

$$m_{\text{low}}(\omega_f) = \varsigma(-\text{sharpness} \cdot (\omega_f - \text{cutoff})), \quad (10)$$

$$m_{\text{high}}(\omega_f) = 1 - m_{\text{low}}(\omega_f). \quad (11)$$

Here m_{low} is a smooth monotone-decreasing low-pass filter and m_{high} is its exact complement, so the masks form a *partition of unity*: $\sum_k m_k(\omega_f) = 1$ for every bin. For general K , we use $K - 1$ ordered cutoffs $0 < c_1 < \dots < c_{K-1} < 1$ (ordering enforced by a cumulative-softplus parameterization) and define

$$\begin{aligned} m_1 &= \varsigma(-\tau_1(\omega - c_1)), \\ m_k &= \varsigma(-\tau_k(\omega - c_k)) - \varsigma(-\tau_{k-1}(\omega - c_{k-1})), \quad 1 < k < K, \\ m_K &= 1 - \varsigma(-\tau_{K-1}(\omega - c_{K-1})), \end{aligned} \tag{12}$$

which again partition unity by construction.

Application to the spectrum.. The masks $m_k \in \mathbb{R}^F$ are real-valued and multiply the complex spectrum bin-wise (the same real scalar scales the real and imaginary parts, so phase is preserved and the gating is pure magnitude):

$$\mathbf{S} = \text{rfft}(\mathbf{s}^n, n=L), \quad \mathbf{S}^{(k)} = m_k \odot \mathbf{S}, \quad \mathbf{s}^{n,(k)} = \text{irfft}(\mathbf{S}^{(k)}, n=L), \tag{13}$$

where $\mathbf{S}, \mathbf{S}^{(k)} \in \mathbb{C}^F$ and $\mathbf{s}^{n,(k)} \in \mathbb{R}^L$. The masks depend only on the learnable scalars (recomputed cheaply each forward pass) and are independent of both the batch and the channel.

Properties.. The decomposition has four properties that make it defensible: (1) *Lossless partition*: since $\sum_k m_k = 1$ exactly, $\sum_k \mathbf{s}^{n,(k)} = \text{irfft}(\sum_k m_k \odot \mathbf{S}) = \text{irfft}(\mathbf{S}) = \mathbf{s}^n$, so reconstruction is exact *before* the heads. (2) *DLinear as a special case*: with $K=2$ and sharpness $\rightarrow \infty$, the soft mask approaches an ideal low/high split, recovering the trend/residual inductive bias of DLinear’s fixed box filter as one member of a strictly richer, learnable family. (3) *Differentiability*: cutoff and sharpness receive gradients through irfft , so the split adapts to each dataset’s spectral content during training. (4) *Cheapness*: the decomposition adds only $2(K - 1)$ scalar parameters, and the FFT/iFFT cost is $O(L \log L)$ per series, negligible relative to the $H \cdot L$ linear heads.

Contrast with FITS.. FITS [10], the closest prior art, applies a low-pass cutoff in the rFFT domain and forecasts with a single complex linear layer, thereby *discarding* all high-frequency content above the cutoff. FREQLITE’s split is different in kind: it is a partition of unity that *keeps* all frequency content and routes each band to its own dedicated head, so the high-frequency band is *modeled* rather than thrown away. This is a falsifiable difference that the ablation in Section 6 tests directly; if the spectral split adds little over RLinear once A-RevIN is present, we report that plainly.

Initialization.. For $K=2$ we initialize cutoff ≈ 0.25 (raw $c = \text{logit}(0.25) \approx -1.0986$), so the low band starts as roughly the lowest quarter of frequencies (a trend / low-season prior), and sharpness ≈ 10 . These are starting points and are subsequently learned; we assert $0.02 < \text{cutoff} < 0.98$ at initialization to avoid a degenerate all-pass or all-stop mask, and we report the final learned values per dataset as an interpretability artifact.

3.4. Per-Band Linear Heads and Recombination

Heads.. Each band is forecast by one independent linear head mapping the lookback length L to the horizon length H :

$$\mathbf{p}^{(k)} = \mathbf{W}_k \mathbf{s}^{n,(k)} + \mathbf{b}_k, \quad \mathbf{W}_k \in \mathbb{R}^{H \times L}, \mathbf{b}_k \in \mathbb{R}^H, \quad (14)$$

realized as a single `nn.Linear(L, H)` with bias. Each head has $H \cdot L + H$ parameters.

Recombination.. The primary model uses an identity sum,

$$\mathbf{p} = \sum_{k=1}^K \mathbf{p}^{(k)} \in \mathbb{R}^H. \quad (15)$$

Because the bands sum to the input (lossless partition) and each head is linear, the identity sum keeps the model in the same hypothesis class as a single linear head while granting the ability to apply a *different* linear map to different frequency content—which is the entire point. In particular, with $K=1$, an all-pass mask, and identity recombination, FREQLITE degenerates exactly to RLinear (RevIN + Linear); this nesting underpins the ablation table. As an ablation we also consider a learnable band gate $\mathbf{g} \in \mathbb{R}^K$ with $\mathbf{p} = \sum_k g_k \mathbf{p}^{(k)}$, reported only if it helps and we can explain why.

3.5. Complexity and Parameter Count

For the primary configuration $K=2$, the total parameter count is

$$\underbrace{2(H \cdot L + H)}_{\text{heads}} + \underbrace{2(K - 1)}_{\text{decomposition}} + \underbrace{3H + 3}_{A-\text{RevIN}} \approx 2HL + 5H + 5. \quad (16)$$

For $L=96, H=96$ this is $\approx 18.9\text{k}$ parameters, and for $L=336, H=720$ it is $\approx 487.5\text{k}$. By comparison, DLinear uses $\approx 2HL$ and RLinear $\approx HL$, so FREQLITE is roughly twice the size of a single linear head—still linear-scale and easily within the 4 GB budget. We report this $2\times$ parameter cost honestly: the claim is competitive or better accuracy at still-linear scale, not a free improvement. The per-forward compute is dominated by the K matrix–vector products of the heads ($O(KHL)$ per series), with the FFT/iFFT adding only $O(L \log L)$; we report exact analytic FLOPs alongside measured train time and peak memory in Section 7.

4. Experimental Setup

A central goal of this work is to show that competitive long-term forecasting is achievable under a strict commodity-hardware budget. Accordingly, every experiment in this paper—training, evaluation, ablation, and efficiency profiling, for FREQLITE and all baselines—is run on a *single* NVIDIA RTX 3050 Ti laptop GPU with 4 GB of VRAM. We re-implement and run all baselines ourselves under identical data splits, normalization, seeds, and hardware, so all comparisons are like-for-like rather than copied from other papers.

Table 1: Summary of the benchmark datasets. C is the number of channels; “Steps” is the total series length. ETT splits follow the standard 12/4/4-month train/validation/test convention; all others use a chronological 70%/10%/20% split.

Dataset	C	Frequency	Steps	Split
ETTh1	7	hourly	17,420	12/4/4 mo
ETTh2	7	hourly	17,420	12/4/4 mo
ETTM1	7	15-min	69,680	12/4/4 mo
ETTM2	7	15-min	69,680	12/4/4 mo
Weather	21	10-min	52,696	70/10/20%
Electricity	321	hourly	26,304	70/10/20%
Traffic	862	hourly	17,544	70/10/20%

4.1. Datasets

We evaluate on seven widely used public long-term forecasting benchmarks: the four Electricity Transformer Temperature datasets (ETTh1, ETTh2, ETTm1, ETTm2), Weather, Electricity (ECL), and Traffic. The ETT datasets record seven power-system variates; ETTh1/ETTh2 are sampled hourly and ETTm1/ETTM2 every 15 minutes. Weather records 21 meteorological variates at 10-minute resolution. Electricity records 321 hourly electricity consumption variates; Traffic records 862 hourly road-occupancy variates from the Bay Area freeway system. The higher-variate datasets (ECL, Traffic) expose the channel-mixing overhead of Transformer-based models. Table 1 summarizes all seven datasets. All series are multivariate and forecast in the channel-independent setting (Section 3.1), except iTransformer and TimesNet which are channel-mixing architectures (see Section 4.3). To probe behavior under distribution shift, we additionally use two datasets outside the standard suite in the non-stationarity study (Section 5.4): Exchange-rate and ILI (national illness), evaluated at the conventional shorter horizons $H \in \{24, 36, 48, 60\}$.

4.2. Forecasting Protocol

We follow the established long-term forecasting protocol [5, 4] so that our numbers are directly comparable to published baselines. For the ETT datasets we use the canonical 12/4/4-month train/validation/test split; Weather uses a chronological 70%/10%/20% split. Each series is z-score normalized using statistics computed on the *training portion only* (per channel), which prevents leakage and matches the standard normalization that makes MSE/MAE comparable across papers. The model’s internal A-RevIN/RevIN then operates as an instance normalization on top of this dataset-level normalization, and the loss and metrics are computed in the train-z-scored space.

We report two lookback lengths: $L=336$, the strong setting for linear models, as the *headline* configuration, and $L=96$ for completeness. For every lookback we forecast the four standard horizons $H \in \{96, 192, 336, 720\}$. We evaluate with Mean Squared Error (MSE) and Mean Absolute Error (MAE), averaged over all horizon steps and

channels. Every (dataset, H , model) cell is run with three seeds {2021, 2022, 2023}, and all tables report the mean $\pm$ standard deviation over seeds.

4.3. Baselines

We compare against a sanity floor and seven competitive forecasters spanning the linear, lightweight-frequency, and Transformer families:

- **Naive (repeat-last)**: predicts the last observed value for the entire horizon; a parameter-free sanity floor.
- **NLinear** [5]: subtracts the last lookback value, applies a single $\text{Linear}(L, H)$, and adds it back—a minimal distribution-shift-robust linear model.
- **DLinear** [5]: moving-average trend/seasonal decomposition with two $\text{Linear}(L, H)$ heads, summed; the time-domain decomposition baseline FREQLITE generalizes.
- **RLinear** [6]: RevIN [7] plus a single $\text{Linear}(L, H)$; the normalization baseline A-RevIN generalizes.
- **FITS** [10]: RevIN plus a low-pass cutoff in the rFFT domain and a single complex linear layer; the closest lightweight frequency-domain prior art, which *discards* the high-frequency band FREQLITE retains.
- **PatchTST-small** [4]: a deliberately small, channel-independent patch Transformer (patch length 16, stride 8, $d_{\text{model}}=64$, 4 heads, 2 layers, dropout 0.2). Run only on the five low- C datasets (ETT + Weather); the attention memory of $O(L^2)$ per patch makes it 4 GB-infeasible on Electricity ($C=321$) and Traffic ($C=862$) at the lookbacks we use.
- **iTransformer** [11]: inverted self-attention over variate tokens (ICLR 2024); each channel is embedded as a token and attention operates across channels rather than time steps. Config: $d_{\text{model}}=128$, 4 heads, 2 encoder layers. A 4 GB-feasible configuration for all seven datasets.
- **TimesNet-small** [12]: rFFT period detection plus 2D Inception convolutions within each dominant period (ICLR 2023), scaled to $d_{\text{model}}=d_{\text{ff}}=32$, $k=3$ dominant periods, 2 layers. As a channel-mixing model its memory scales moderately with C .

iTransformer and TimesNet are channel-mixing architectures; we run them on all seven datasets with the same protocol but dataset-specific batch sizes (16 for Weather and Electricity, 8 for Traffic) to remain within the 4 GB budget.

4.4. Implementation Details

All models are implemented in PyTorch 2.6.0 (CUDA 12.4) and trained on a single RTX 3050 Ti laptop GPU (4 GB). We use the Adam optimizer [19] with learning rate 10^{-3} , no weight decay, and MSE loss. Training uses a batch size of 32, at most 20 epochs with early stopping (patience 3 on the validation loss), a type-1 step learning-rate

Table 2: Long-term forecasting results (lookback $L = 336$). Each cell is test MSE/MAE (mean $\pm$ std over 3 seeds) in the train-standardized space. **Bold**: best; underline: second best. Standard benchmarks. PatchTST* uses a 4 GB-feasible small config.

Dataset	H	Naive		NLinear		DLinear		RLinear		FiTS		PatchTST*		iTransformer		TimesNet		FreqLite	
		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	96	1.294 $\pm$ 0.000	0.713 $\pm$ 0.000	0.384 $\pm$ 0.005	0.405 $\pm$ 0.004	<u>0.376$\pm$0.001</u>	<u>0.398$\pm$0.002</u>	0.379 $\pm$ 0.003	0.400 $\pm$ 0.003	0.399 $\pm$ 0.000	0.419 $\pm$ 0.000	0.431 $\pm$ 0.043	0.434 $\pm$ 0.024	0.403 $\pm$ 0.002	0.416 $\pm$ 0.002	0.457 $\pm$ 0.051	0.454 $\pm$ 0.020	0.373$\pm$0.001	0.395$\pm$0.001
	192	1.325 $\pm$ 0.000	0.733 $\pm$ 0.000	0.413 $\pm$ 0.001	0.421 $\pm$ 0.000	0.418 $\pm$ 0.009	0.428 $\pm$ 0.009	<u>0.412$\pm$0.001</u>	<u>0.419$\pm$0.001</u>	0.429 $\pm$ 0.000	0.436 $\pm$ 0.000	0.444 $\pm$ 0.006	0.445 $\pm$ 0.005	0.447 $\pm$ 0.010	0.445 $\pm$ 0.007	0.530 $\pm$ 0.035	0.499 $\pm$ 0.017	0.410$\pm$0.005	0.417$\pm$0.005
	336	1.330 $\pm$ 0.000	0.746 $\pm$ 0.000	<u>0.438$\pm$0.002</u>	<u>0.437$\pm$0.002</u>	0.453 $\pm$ 0.011	0.454 $\pm$ 0.011	0.442 $\pm$ 0.007	0.440 $\pm$ 0.006	0.451 $\pm$ 0.000	0.448 $\pm$ 0.000	0.552 $\pm$ 0.058	0.503 $\pm$ 0.027	0.472 $\pm$ 0.017	0.461 $\pm$ 0.010	0.600 $\pm$ 0.106	0.539 $\pm$ 0.047	0.432$\pm$0.001	0.430$\pm$0.001
	720	1.335 $\pm$ 0.000	0.755 $\pm$ 0.000	<u>0.444$\pm$0.000</u>	<u>0.459$\pm$0.000</u>	0.488 $\pm$ 0.016	0.501 $\pm$ 0.012	0.449 $\pm$ 0.002	0.463 $\pm$ 0.002	0.447 $\pm$ 0.001	0.466 $\pm$ 0.001	0.630 $\pm$ 0.029	0.557 $\pm$ 0.014	0.485 $\pm$ 0.011	0.489 $\pm$ 0.007	0.673 $\pm$ 0.031	0.577 $\pm$ 0.015	0.444$\pm$0.002	0.459$\pm$0.002
ETTh2	96	0.432 $\pm$ 0.000	0.422 $\pm$ 0.000	0.283 $\pm$ 0.001	0.344 $\pm$ 0.001	0.285 $\pm$ 0.003	0.350 $\pm$ 0.003	<u>0.281$\pm$0.000</u>	<u>0.342$\pm$0.000</u>	0.285 $\pm$ 0.001	0.346 $\pm$ 0.000	0.346 $\pm$ 0.016	0.382 $\pm$ 0.008	0.322 $\pm$ 0.015	0.371 $\pm$ 0.006	0.406 $\pm$ 0.024	0.427 $\pm$ 0.007	0.275$\pm$0.001	0.336$\pm$0.000
	192	0.534 $\pm$ 0.000	0.473 $\pm$ 0.000	0.347 $\pm$ 0.003	0.386 $\pm$ 0.001	0.385 $\pm$ 0.009	0.420 $\pm$ 0.005	0.349 $\pm$ 0.013	0.386 $\pm$ 0.006	<u>0.344$\pm$0.000</u>	<u>0.383$\pm$0.000</u>	0.423 $\pm$ 0.029	0.429 $\pm$ 0.013	0.401 $\pm$ 0.018	0.418 $\pm$ 0.009	0.425 $\pm$ 0.023	0.441 $\pm$ 0.010	0.340$\pm$0.005	0.380$\pm$0.003
	336	0.597 $\pm$ 0.000	0.511 $\pm$ 0.000	0.384 $\pm$ 0.013	0.415 $\pm$ 0.006	0.457 $\pm$ 0.007	0.469 $\pm$ 0.005	0.369 $\pm$ 0.004	0.407 $\pm$ 0.002	<u>0.364$\pm$0.000</u>	0.401$\pm$0.000	0.432 $\pm$ 0.006	0.440 $\pm$ 0.003	0.410 $\pm$ 0.003	0.430 $\pm$ 0.001	0.431 $\pm$ 0.015	0.447 $\pm$ 0.012	0.362$\pm$0.001	<u>0.402$\pm$0.000</u>
	720	0.594 $\pm$ 0.000	0.519 $\pm$ 0.000	0.407 $\pm$ 0.007	0.442 $\pm$ 0.002	0.701 $\pm$ 0.027	0.594 $\pm$ 0.012	0.397 $\pm$ 0.001	0.433 $\pm$ 0.000	0.391$\pm$0.000	0.427$\pm$0.000	0.421 $\pm$ 0.016	0.447 $\pm$ 0.008	0.441 $\pm$ 0.005	0.458 $\pm$ 0.001	0.479 $\pm$ 0.006	0.481 $\pm$ 0.001	<u>0.392$\pm$0.002</u>	<u>0.430$\pm$0.001</u>
ETTm1	96	1.214 $\pm$ 0.000	0.665 $\pm$ 0.000	0.305 $\pm$ 0.002	0.347 $\pm$ 0.002	0.300$\pm$0.001	<u>0.344$\pm$0.000</u>	0.305 $\pm$ 0.004	0.346 $\pm$ 0.003	0.304 $\pm$ 0.000	0.346 $\pm$ 0.000	0.302 $\pm$ 0.013	0.352 $\pm$ 0.008	0.323 $\pm$ 0.010	0.364 $\pm$ 0.001	0.380 $\pm$ 0.029	0.397 $\pm$ 0.013	<u>0.302$\pm$0.001</u>	0.344$\pm$0.001
	192	1.261 $\pm$ 0.000	0.690 $\pm$ 0.000	0.343 $\pm$ 0.003	0.370 $\pm$ 0.003	0.337$\pm$0.002	0.368 $\pm$ 0.003	0.338 $\pm$ 0.003	<u>0.366$\pm$0.003</u>	<u>0.338$\pm$0.000</u>	0.366$\pm$0.000	0.345 $\pm$ 0.001	0.381 $\pm$ 0.003	0.356 $\pm$ 0.007	0.385 $\pm$ 0.003	0.472 $\pm$ 0.040	0.441 $\pm$ 0.021	0.340 $\pm$ 0.004	0.367 $\pm$ 0.003
	336	1.287 $\pm$ 0.000	0.707 $\pm$ 0.000	0.374 $\pm$ 0.002	0.386 $\pm$ 0.001	0.372$\pm$0.001	0.390 $\pm$ 0.001	0.373 $\pm$ 0.001	0.386 $\pm$ 0.001	0.372 $\pm$ 0.000	<u>0.385$\pm$0.000</u>	0.381 $\pm$ 0.011	0.401 $\pm$ 0.006	0.385 $\pm$ 0.005	0.405 $\pm$ 0.002	0.454 $\pm$ 0.007	0.443 $\pm$ 0.002	<u>0.372$\pm$0.001</u>	0.384$\pm$0.001
	720	1.322 $\pm$ 0.000	0.729 $\pm$ 0.000	0.431 $\pm$ 0.003	0.419 $\pm$ 0.002	<u>0.428$\pm$0.004</u>	0.424 $\pm$ 0.006	0.431 $\pm$ 0.003	0.418 $\pm$ 0.001	0.428$\pm$0.000	0.416$\pm$0.000	0.429 $\pm$ 0.007	0.436 $\pm$ 0.006	0.441 $\pm$ 0.002	0.435 $\pm$ 0.002	0.520 $\pm$ 0.044	0.478 $\pm$ 0.015	0.428 $\pm$ 0.001	<u>0.417$\pm$0.001</u>
ETTm2	96	0.266 $\pm$ 0.000	0.328 $\pm$ 0.000	0.167 $\pm$ 0.001	0.257 $\pm$ 0.000	0.169 $\pm$ 0.002	0.265 $\pm$ 0.002	<u>0.166$\pm$0.000</u>	<u>0.255$\pm$0.000</u>	0.167 $\pm$ 0.000	0.256 $\pm$ 0.000	0.177 $\pm$ 0.002	0.262 $\pm$ 0.001	0.187 $\pm$ 0.005	0.275 $\pm$ 0.005	0.201 $\pm$ 0.004	0.280 $\pm$ 0.003	0.164$\pm$0.001	0.253$\pm$0.000
	192	0.340 $\pm$ 0.000	0.371 $\pm$ 0.000	0.222 $\pm$ 0.000	0.294 $\pm$ 0.000	0.229 $\pm$ 0.006	0.308 $\pm$ 0.009	<u>0.221$\pm$0.000</u>	<u>0.292$\pm$0.000</u>	0.222 $\pm$ 0.000	0.293 $\pm$ 0.000	0.243 $\pm$ 0.009	0.308 $\pm$ 0.005	0.250 $\pm$ 0.009	0.317 $\pm$ 0.006	0.267 $\pm$ 0.011	0.325 $\pm$ 0.007	0.219$\pm$0.000	0.290$\pm$0.000
	336	0.412 $\pm$ 0.000	0.410 $\pm$ 0.000	0.276 $\pm$ 0.000	0.329 $\pm$ 0.000	0.304 $\pm$ 0.023	0.362 $\pm$ 0.022	<u>0.275$\pm$0.000</u>	0.327 $\pm$ 0.000	0.275 $\pm$ 0.000	<u>0.327$\pm$0.000</u>	0.310 $\pm$ 0.009	0.350 $\pm$ 0.006	0.302 $\pm$ 0.008	0.349 $\pm$ 0.004	0.309 $\pm$ 0.017	0.350 $\pm$ 0.010	0.274$\pm$0.001	0.326$\pm$0.000
	720	0.522 $\pm$ 0.000	0.466 $\pm$ 0.000	0.371 $\pm$ 0.000	0.386 $\pm$ 0.000	0.431 $\pm$ 0.017	0.443 $\pm$ 0.010	<u>0.369$\pm$0.000</u>	0.384 $\pm$ 0.000	0.368$\pm$0.000	0.383$\pm$0.000	0.387 $\pm$ 0.007	0.403 $\pm$ 0.007	0.396 $\pm$ 0.006	0.406 $\pm$ 0.005	0.428 $\pm$ 0.028	0.418 $\pm$ 0.015	0.370 $\pm$ 0.004	<u>0.383$\pm$0.001</u>
Weather	96	0.259 $\pm$ 0.000	0.254 $\pm$ 0.000	0.174 $\pm$ 0.001	0.224 $\pm$ 0.002	0.174 $\pm$ 0.000	0.235 $\pm$ 0.001	0.175 $\pm$ 0.001	0.224 $\pm$ 0.001	0.176 $\pm$ 0.000	0.228 $\pm$ 0.000	0.149$\pm$0.001	0.197$\pm$0.001	<u>0.160$\pm$0.002</u>	<u>0.210$\pm$0.001</u>	0.183 $\pm$ 0.015	0.235 $\pm$ 0.012	0.174 $\pm$ 0.000	0.224 $\pm$ 0.001
	192	0.309 $\pm$ 0.000	0.292 $\pm$ 0.000	0.217 $\pm$ 0.000	0.260 $\pm$ 0.000	0.217 $\pm$ 0.001	0.275 $\pm$ 0.002	0.218 $\pm$ 0.000	0.260 $\pm$ 0.000	0.219 $\pm$ 0.000	0.263 $\pm$ 0.000	0.195$\pm$0.002	0.241$\pm$0.001	<u>0.206$\pm$0.000</u>	<u>0.252$\pm$0.001</u>	0.250 $\pm$ 0.028	0.289 $\pm$ 0.019	0.218 $\pm$ 0.000	0.260 $\pm$ 0.001
	336	0.376 $\pm$ 0.000	0.338 $\pm$ 0.000	0.266 $\pm$ 0.000	0.296 $\pm$ 0.000	0.262 $\pm$ 0.001	0.313 $\pm$ 0.003	0.265 $\pm$ 0.000	0.295 $\pm$ 0.000	0.267 $\pm$ 0.000	0.297 $\pm$ 0.000	0.248$\pm$0.002	0.283$\pm$0.004	<u>0.256$\pm$0.002</u>	<u>0.289$\pm$0.000</u>	0.332 $\pm$ 0.003	0.341 $\pm$ 0.003	0.265 $\pm$ 0.000	0.295 $\pm$ 0.000
	720	0.465 $\pm$ 0.000	0.394 $\pm$ 0.000	0.334 $\pm$ 0.000	0.344 $\pm$ 0.000	0.333 $\pm$ 0.007	0.376 $\pm$ 0.009	0.333 $\pm$ 0.000	0.342 $\pm$ 0.000	0.334 $\pm$ 0.000	0.343 $\pm$ 0.000	<u>0.329$\pm$0.008</u>	<u>0.339$\pm$0.006</u>	0.327$\pm$0.003	0.339$\pm$0.002	0.400 $\pm$ 0.007	0.387 $\pm$ 0.004	0.333 $\pm$ 0.000	0.342 $\pm$ 0.000

schedule (halving the rate each epoch after the first), and global gradient-norm clipping at 1.0. We train in full `fp32` precision: the models are tiny, `fp32` fits comfortably within 4 GB, and it avoids the reproducibility noise of mixed precision. For reproducibility we fix the `torch`, `numpy`, and `random` seeds, enable deterministic algorithms, and disable `cuDNN` autotuning. At each run we record parameters, analytic FLOPs, wall-clock training time per epoch, and peak GPU memory via `torch.cuda.max_memory_allocated`; these populate the efficiency analysis in Section 7. Every reported number is regenerable by a script in the repository.

5. Results

We first present the main long-term forecasting comparison across seven datasets against eight baselines including two recent ICLR Transformer models, then analyze distribution-shift robustness and non-stationary regimes. All accuracy numbers are test MSE and MAE in the train-standardized space, averaged over horizon steps and channels, reported as mean $\pm$ std over three seeds; the best result in each row is in bold and the second best underlined.

5.1. Main Comparison

Tables 2 and 4 report the headline comparisons at $L=336$ and $L=96$ on the five standard benchmarks; Table 3 extends the $L=336$ comparison to the two high-variate datasets (Electricity and Traffic).

At the headline $L=336$ setting, FREQLITE attains the best overall average test MSE of 0.3177 across all seven datasets (vs. RLinear 0.3199, iTransformer 0.3318, TimesNet 0.4134) and the most per-setting wins of any model (10 of the 28 dataset $\times$ horizon settings, vs. 8 for iTransformer, its closest rival). On the five standard low- C benchmarks (ETT + Weather), FREQLITE averages 0.3244 and *surpasses both recent ICLR Transformer baselines*: it beats iTransformer (ICLR 2024, avg MSE 0.3485) by **6.9%** and TimesNet (ICLR 2023, avg MSE 0.4098) by **20.8%**, while also

Table 3: Long-term forecasting results (lookback $L = 336$). Each cell is test MSE/MAE (mean $\pm$ std over 3 seeds) in the train-standardized space. **Bold**: best; underline: second best. Electricity ($C=321$) and Traffic ($C=862$). PatchTST* omitted (4 GB-infeasible at these channel counts).

Dataset	H	Naive		NLinear		DLinear		RLinear		FITS		PatchTST*		iTransformer		TimesNet		FreqLite	
		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
Electricity	96	1.588 $\pm$ 0.000	0.945 $\pm$ 0.000	0.141 $\pm$ 0.000	0.237 $\pm$ 0.000	<u>0.140$\pm$0.000</u>	0.238 $\pm$ 0.000	0.141 $\pm$ 0.000	0.236 $\pm$ 0.000	0.188 $\pm$ 0.000	0.301 $\pm$ 0.000	–	–	0.134$\pm$0.000	0.230$\pm$0.000	0.191 $\pm$ 0.008	0.296 $\pm$ 0.007	0.140 $\pm$ 0.000	<u>0.235$\pm$0.000</u>
	192	1.596 $\pm$ 0.000	0.951 $\pm$ 0.000	0.156 $\pm$ 0.000	0.250 $\pm$ 0.000	<u>0.154$\pm$0.000</u>	0.251 $\pm$ 0.000	0.155 $\pm$ 0.000	<u>0.248$\pm$0.000</u>	0.201 $\pm$ 0.000	0.310 $\pm$ 0.000	–	–	0.153$\pm$0.000	0.248 $\pm$ 0.000	0.205 $\pm$ 0.004	0.306 $\pm$ 0.003	0.154 $\pm$ 0.000	0.247$\pm$0.000
	336	1.618 $\pm$ 0.000	0.961 $\pm$ 0.000	0.172 $\pm$ 0.000	0.266 $\pm$ 0.000	<u>0.170$\pm$0.000</u>	0.268 $\pm$ 0.000	0.172 $\pm$ 0.000	0.265 $\pm$ 0.000	0.216 $\pm$ 0.000	0.323 $\pm$ 0.000	–	–	0.169$\pm$0.001	<u>0.264$\pm$0.001</u>	0.207 $\pm$ 0.000	0.311 $\pm$ 0.001	0.171 $\pm$ 0.000	0.264$\pm$0.000
	720	1.647 $\pm$ 0.000	0.975 $\pm$ 0.000	0.211 $\pm$ 0.000	0.299 $\pm$ 0.000	0.204$\pm$0.000	0.301 $\pm$ 0.000	0.210 $\pm$ 0.000	<u>0.297$\pm$0.000</u>	0.252 $\pm$ 0.000	0.349 $\pm$ 0.000	–	–	<u>0.208$\pm$0.002</u>	0.298 $\pm$ 0.003	0.231 $\pm$ 0.001	0.327 $\pm$ 0.004	0.210 $\pm$ 0.000	0.296$\pm$0.000
Traffic	96	2.714 $\pm$ 0.000	1.077 $\pm$ 0.000	0.411 $\pm$ 0.000	0.281 $\pm$ 0.000	0.410 $\pm$ 0.000	0.283 $\pm$ 0.000	0.410 $\pm$ 0.000	0.279 $\pm$ 0.000	0.520 $\pm$ 0.000	0.385 $\pm$ 0.000	–	–	0.381$\pm$0.002	0.275$\pm$0.000	0.611 $\pm$ 0.010	0.331 $\pm$ 0.004	<u>0.410$\pm$0.000</u>	<u>0.278$\pm$0.000</u>
	192	2.747 $\pm$ 0.000	1.085 $\pm$ 0.000	0.424 $\pm$ 0.000	0.286 $\pm$ 0.000	0.423 $\pm$ 0.000	0.288 $\pm$ 0.000	0.423 $\pm$ 0.000	<u>0.283$\pm$0.000</u>	0.532 $\pm$ 0.000	0.389 $\pm$ 0.000	–	–	0.406$\pm$0.000	0.288 $\pm$ 0.000	0.632 $\pm$ 0.015	0.341 $\pm$ 0.008	<u>0.423$\pm$0.000</u>	0.283$\pm$0.000
	336	2.788 $\pm$ 0.000	1.094 $\pm$ 0.000	0.437 $\pm$ 0.000	0.293 $\pm$ 0.000	0.436 $\pm$ 0.000	0.296 $\pm$ 0.000	0.436 $\pm$ 0.000	<u>0.290$\pm$0.000</u>	0.543 $\pm$ 0.000	0.393 $\pm$ 0.000	–	–	0.421$\pm$0.000	0.295 $\pm$ 0.000	0.634 $\pm$ 0.003	0.341 $\pm$ 0.004	<u>0.435$\pm$0.000</u>	0.289$\pm$0.000
	720	2.809 $\pm$ 0.000	1.097 $\pm$ 0.000	0.465 $\pm$ 0.000	0.310 $\pm$ 0.000	0.466 $\pm$ 0.000	0.315 $\pm$ 0.000	0.464 $\pm$ 0.000	<u>0.306$\pm$0.000</u>	0.564 $\pm$ 0.000	0.403 $\pm$ 0.000	–	–	0.450$\pm$0.000	0.311 $\pm$ 0.000	0.667 $\pm$ 0.003	0.356 $\pm$ 0.002	<u>0.463$\pm$0.000</u>	0.306$\pm$0.000

Table 4: Long-term forecasting results (lookback $L = 96$). Each cell is test MSE/MAE (mean $\pm$ std over 3 seeds) in the train-standardized space. **Bold**: best; underline: second best. Standard benchmarks. PatchTST* uses a 4 GB-feasible small config.

Dataset	H	Naive		NLinear		DLinear		RLinear		FITS		PatchTST*		iTransformer		TimesNet		FreqLite	
		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	96	1.294±0.000	0.713±0.000	0.398±0.003	0.407±0.003	0.390±0.003	0.404±0.005	0.391±0.000	0.399±0.000	0.409±0.001	0.417±0.000	0.376±0.003	<u>0.396±0.000</u>	0.387±0.001	0.405±0.001	0.401±0.003	0.416±0.003	<u>0.386±0.000</u>	0.394±0.000
	192	1.325±0.000	0.733±0.000	0.446±0.001	0.433±0.001	0.440±0.003	0.434±0.004	0.443±0.001	<u>0.429±0.001</u>	0.459±0.001	0.446±0.001	0.442±0.000	0.436±0.003	0.434±0.003	0.432±0.002	0.485±0.021	0.471±0.011	<u>0.437±0.000</u>	0.423±0.000
	336	1.330±0.000	0.746±0.000	0.488±0.001	0.454±0.001	0.487±0.004	0.464±0.004	<u>0.486±0.000</u>	<u>0.450±0.000</u>	0.504±0.005	0.470±0.005	0.507±0.003	0.469±0.002	0.486±0.003	0.461±0.001	0.519±0.037	0.485±0.016	0.481±0.000	0.446±0.000
	720	1.335±0.000	0.755±0.000	0.482±0.000	<u>0.472±0.000</u>	0.510±0.002	0.505±0.002	0.487±0.000	0.474±0.000	0.496±0.007	0.488±0.005	0.529±0.022	0.500±0.007	0.481±0.010	0.478±0.007	0.575±0.046	0.523±0.018	<u>0.482±0.000</u>	0.470±0.000
ETTh2	96	0.432±0.000	0.422±0.000	0.298±0.000	0.347±0.000	0.327±0.002	0.382±0.003	<u>0.295±0.000</u>	<u>0.344±0.000</u>	0.303±0.000	0.349±0.001	0.298±0.007	0.346±0.004	0.316±0.002	0.358±0.003	0.335±0.018	0.367±0.009	0.290±0.000	0.339±0.000
	192	0.534±0.000	0.473±0.000	0.385±0.000	0.399±0.000	0.450±0.005	0.457±0.003	0.382±0.000	0.396±0.000	0.387±0.001	0.398±0.001	0.376±0.006	<u>0.394±0.004</u>	0.381±0.004	0.400±0.002	0.419±0.016	0.417±0.003	<u>0.376±0.000</u>	0.391±0.000
	336	0.597±0.000	0.511±0.000	0.425±0.001	0.434±0.000	0.567±0.010	0.527±0.005	<u>0.421±0.000</u>	<u>0.431±0.000</u>	0.424±0.001	0.432±0.000	0.424±0.001	0.432±0.002	0.427±0.003	0.434±0.001	0.445±0.009	0.441±0.002	0.416±0.000	0.427±0.000
	720	0.594±0.000	0.519±0.000	0.430±0.000	0.450±0.000	0.781±0.027	0.635±0.012	0.427±0.000	0.445±0.000	0.429±0.000	0.444±0.001	<u>0.425±0.009</u>	<u>0.443±0.005</u>	0.431±0.001	0.447±0.001	0.441±0.005	0.453±0.003	0.424±0.002	0.442±0.001
ETTh1	96	1.214±0.000	0.665±0.000	0.349±0.002	0.371±0.002	0.344±0.002	0.372±0.003	0.352±0.001	0.373±0.001	0.362±0.000	0.381±0.000	0.324±0.004	0.364±0.003	<u>0.332±0.001</u>	<u>0.366±0.001</u>	0.344±0.013	0.375±0.006	0.351±0.002	0.372±0.000
	192	1.261±0.000	0.690±0.000	0.391±0.000	0.391±0.000	<u>0.381±0.001</u>	0.391±0.000	0.391±0.001	0.392±0.001	0.397±0.000	0.397±0.000	0.369±0.003	0.391±0.002	0.381±0.005	0.393±0.002	0.402±0.008	0.406±0.003	0.389±0.002	<u>0.391±0.002</u>
	336	1.287±0.000	0.707±0.000	0.424±0.002	<u>0.413±0.001</u>	<u>0.413±0.000</u>	0.413±0.001	0.424±0.001	0.414±0.001	0.429±0.000	0.417±0.000	0.391±0.001	0.407±0.001	0.424±0.004	0.417±0.002	0.454±0.008	0.435±0.006	0.423±0.001	0.413±0.001
	720	1.322±0.000	0.729±0.000	0.486±0.000	<u>0.448±0.000</u>	<u>0.474±0.003</u>	0.453±0.005	0.487±0.000	0.449±0.000	0.489±0.000	0.450±0.000	0.451±0.002	0.442±0.000	0.492±0.012	0.456±0.005	0.542±0.035	0.479±0.012	0.487±0.002	0.449±0.000
ETTh2	96	0.266±0.000	0.328±0.000	0.183±0.000	0.266±0.001	0.191±0.002	0.287±0.004	0.183±0.000	0.266±0.000	0.186±0.000	0.269±0.000	0.177±0.001	0.261±0.000	0.189±0.004	0.272±0.005	0.189±0.007	0.267±0.004	<u>0.182±0.000</u>	<u>0.264±0.001</u>
	192	0.340±0.000	0.371±0.000	0.247±0.000	0.305±0.001	0.273±0.007	0.348±0.010	0.247±0.000	0.249±0.000	0.307±0.000	0.245±0.003	0.304±0.003	0.252±0.001	0.311±0.001	0.259±0.005	0.311±0.004	<u>0.246±0.000</u>	<u>0.304±0.000</u>	
	336	0.412±0.000	0.410±0.000	0.309±0.000	0.344±0.000	0.348±0.015	0.400±0.013	<u>0.308±0.000</u>	0.344±0.000	0.309±0.000	<u>0.344±0.000</u>	0.310±0.005	0.347±0.005	0.317±0.003	0.352±0.002	0.324±0.018	0.350±0.009	0.308±0.000	0.343±0.000
	720	0.522±0.000	0.466±0.000	0.409±0.000	0.400±0.000	0.527±0.053	0.505±0.032	0.409±0.000	0.399±0.000	0.409±0.000	<u>0.399±0.000</u>	0.407±0.004	0.405±0.003	0.417±0.000	0.407±0.001	0.440±0.008	0.414±0.004	<u>0.408±0.000</u>	0.398±0.000
Weather	96	0.259±0.000	0.254±0.000	0.195±0.000	0.234±0.000	0.197±0.002	0.257±0.004	0.195±0.000	0.234±0.000	0.197±0.000	0.238±0.000	<u>0.175±0.001</u>	<u>0.217±0.001</u>	0.176±0.005	0.216±0.006	0.169±0.002	0.217±0.002	0.195±0.000	0.234±0.000
	192	0.309±0.000	0.292±0.000	0.241±0.000	0.271±0.000	0.244±0.007	0.306±0.010	0.240±0.000	0.270±0.000	0.242±0.000	0.273±0.000	<u>0.221±0.000</u>	<u>0.258±0.000</u>	0.222±0.002	0.256±0.001	0.220±0.004	0.261±0.003	0.240±0.000	0.270±0.000
	336	0.376±0.000	0.338±0.000	0.293±0.000	0.308±0.000	0.285±0.002	0.336±0.002	0.291±0.000	0.306±0.000	0.294±0.000	0.309±0.000	0.278±0.001	0.299±0.001	0.280±0.001	0.298±0.000	0.284±0.006	0.306±0.005	0.292±0.000	0.306±0.000
	720	0.465±0.000	0.394±0.000	0.366±0.000	0.356±0.000	0.350±0.004	0.387±0.006	0.364±0.000	0.353±0.000	0.366±0.000	0.355±0.000	0.355±0.001	<u>0.249±0.001</u>	<u>0.294±0.001</u>	0.355±0.001	0.349±0.001	0.371±0.011	0.360±0.007	0.364±0.001

reduces the advantage of frequency decomposition. Detailed $L=96$ analysis follows the table.

5.2. Discussion

Three findings stand out.

(1) Transformers underperform at long lookback—except where channel mixing pays. Both ICLR Transformers trail FREQLITE at $L=336$ overall: iTransformer by 4.3% and TimesNet by 23.2% on paired cells across the seven datasets (6.9% and 20.8% on the standard five). TimesNet’s 2D convolutions over detected periods are expensive (1047 MB peak GPU memory, 93 s/epoch on average, Section 7) yet never translate to better accuracy at long lookback. iTransformer is the more serious competitor: its inverted variate-token attention is memory-efficient (79 MB) and wins on Traffic, where 862 strongly correlated sensors reward channel mixing. Everywhere else the channel-independent FREQLITE matches or beats it at a fraction of the training cost.

(2) FREQLITE vs. the linear family. On largely stationary benchmarks, the accuracy gap over the best linear model (RLinear) remains modest—0.7% mean paired improvement at $L=336$ over all seven datasets—but it is statistically robust rather than noise: a paired Wilcoxon signed-rank test across all 84 matched (dataset, horizon, seed) cells gives $p \approx 1.8 \times 10^{-9}$ (Table 5), and FREQLITE improves significantly over every linear and lightweight-frequency baseline at both lookbacks ($p < 10^{-3}$ in all such comparisons). The gap over iTransformer is also highly significant ($p \approx 2 \times 10^{-5}$ at $L=336$).

(3) Short context. At $L=96$, PatchTST-small is the overall best on the standard five datasets (0.3541 avg MSE, 10 wins): with a short context the Transformer’s representational capacity pays off. FREQLITE is second (0.3589, 5 wins), statistically tied with iTransformer (0.3590; paired difference +0.03%, ns) and ahead of RLinear (0.3611), NLinear (0.3623), FITS (0.3669), TimesNet (0.3810), and DLinear (0.3989). The headline regime for a linear-scale forecaster is the long-lookback setting, and there FREQLITE leads all eight baselines in overall average MSE.

5.3. Distribution-Shift Stress Test

A-RevIN’s claimed advantage over standard RevIN is robustness when the test distribution differs from training. To test this *directly*, we stratify test windows by a shift score—the mean absolute z-scored input mean, measuring how far each window’s input has drifted from the training distribution—and compare models on the Near, Mid, and Far thirds (Table 6). We run this protocol on ETTh1, ETTm2, Weather, and Exchange-rate at $L=336$, $H=96$, using three seeds each.

The results reveal a nuanced picture. Across all four datasets and all three shift thirds, FREQLITE consistently beats RLinear (by 1.4–1.6% on average), including in the Far third where shift is highest. This shows that the frequency decomposition provides consistent gains *irrespective* of distribution shift. However, isolating A-RevIN’s contribution by comparing A-RevIN-Linear (identical to RLinear except for the adaptive normalization gate) against RLinear shows near-zero differences on these datasets (ratios 0.998–1.000 $\times$): the standard benchmarks have low-to-moderate non-stationarity after z-score normalization, so the adaptive gate rarely engages. A-RevIN’s targeted

Table 5: Paired significance tests for FREQLITE versus each baseline (Wilcoxon signed-rank over all matched (dataset, horizon, seed) cells). Δ MSE is the mean relative improvement of FREQLITE (positive = FREQLITE better); n is the number of paired cells. *** $p < 10^{-3}$, ** $p < 10^{-2}$, * $p < 0.05$, ns = not significant.

Setting	Baseline	n	Δ MSE (%)	p (Wilcoxon)
$L=336$	RLINEAR	84	+0.69	1.8e-09 (***)
$L=336$	NLINEAR	84	+1.14	1.5e-11 (***)
$L=336$	DLINEAR	84	+6.57	3.6e-05 (***)
$L=336$	FITS	84	+7.33	4.8e-12 (***)
$L=336$	PatchTST	60	+9.57	8.7e-06 (***)
$L=336$	iTransformer	84	+4.25	2.1e-05 (***)
$L=336$	TimesNet	84	+23.15	1.8e-15 (***)
$L=96$	RLINEAR	60	+0.60	7.4e-08 (***)
$L=96$	NLINEAR	60	+0.94	2.5e-08 (***)
$L=96$	DLINEAR	60	+10.04	5.4e-04 (***)
$L=96$	FITS	60	+2.17	1.7e-11 (***)
$L=96$	PatchTST	60	-1.36	1.4e-02 (*)
$L=96$	iTransformer	60	+0.03	8.8e-01 (ns)
$L=96$	TimesNet	60	+5.80	2.2e-06 (***)
ILI	RLINEAR	12	+3.90	4.9e-04 (***)

advantage activates on genuinely non-stationary data—confirmed in Section 5.4 on ILI, where it reduces MSE by 0.38–0.66% in isolation and FreqLite improves over RLinear by 3.2–4.8%. Taken together, the frequency decomposition provides universal gains and A-RevIN adds targeted robustness for non-stationary distributions.

5.4. Non-stationary Regimes and the Role of A-RevIN

A-RevIN is designed to engage under non-stationarity and to reduce to RevIN otherwise. To test this directly we evaluate two datasets outside the standard suite: Exchange-rate, which is only mildly non-stationary, and ILI (national illness), which is strongly non-stationary. Table 7 reports MSE on both.

On the mildly non-stationary Exchange-rate data, DLinear is the best model and A-RevIN is neutral-to-slightly-negative: isolating its effect by comparing RLinear with an A-RevIN-Linear of the same backbone gives differences of only +0.000, −0.000, −0.001, and −0.002 across the four horizons. We report honestly that FREQLITE is competitive *within* the RevIN family here but that DLinear’s time-domain decomposition wins on this dataset.

On the strongly non-stationary ILI data, A-RevIN clearly helps. Isolating its effect on the same single-head backbone (RLinear vs. A-RevIN-Linear), the adaptive normalization reduces MSE by 0.66%, 0.63%, 0.52%, and

Table 6: Distribution-shift stress test. Test windows ranked by shift score (z-scored input mean), split into Near/Mid/Far thirds. Mean test MSE over three seeds. **Bold**: best per row.

Dataset	Shift third	RLinear	A-RevIN-Linear	FreqLite
ETTh1	Near	0.318	0.318	0.313
	Mid	0.360	0.359	0.354
	Far	0.459	0.459	0.453
	<i>Far–Near gap</i>	+0.141	+0.142	+0.140
ETTh2	Near	0.186	0.185	0.184
	Mid	0.130	0.130	0.128
	Far	0.182	0.182	0.180
	<i>Far–Near gap</i>	-0.004	-0.003	-0.004
Weather	Near	0.207	0.207	0.206
	Mid	0.152	0.152	0.152
	Far	0.165	0.165	0.164
	<i>Far–Near gap</i>	-0.042	-0.042	-0.042
Exchange	Near	0.074	0.074	0.073
	Mid	0.083	0.083	0.083
	Far	0.096	0.096	0.095
	<i>Far–Near gap</i>	+0.022	+0.022	+0.022

0.38% at horizons $H \in \{24, 36, 48, 60\}$. The full FREQ-LITE improves over RLinear by 4.0%, 4.8%, 3.5%, and 3.2% (mean 3.9%) across the same horizons, while DLinear and FITS are far worse on ILI (MSE in the 4.6–6.6 range versus FREQ-LITE’s 3.3–3.9).

Gate sweep and a gradient-trap finding. Table 8 sweeps the A-RevIN gate on ILI. As the learned mean gate $\bar{\rho}$ increases from 0.02 (with raw gate initialization $\rho_0 = -4$) through 0.51 ($\rho_0 = 0$) to 1.00 (forced), the ILI error decreases *monotonically*. This exposes a methodological subtlety: with $\rho_0 = -4$ the gate is gradient-starved. Because the adaptive parameters are initialized at their identity values ($\mathbf{a} = \mathbf{b} = \boldsymbol{\lambda} = \mathbf{0}$), the gradient $\partial \mathcal{L} / \partial \rho$ is approximately zero at initialization, so the gate never opens and A-RevIN stays at RevIN even on data where adapting would help (the gate remained at $\bar{\rho} = 0.02$ on ILI). Initializing $\rho_0 = 0$, now our default, lets the gate engage, while RevIN is still recovered exactly at $\rho = 0$. This is the recommended initialization for A-RevIN.

Interpretability. Figure 2 shows the learned spectral mask, and Figure 3 the per-step A-RevIN correction profile, giving direct evidence that the components behave as designed rather than merely adding parameters.

Table 7: Non-stationarity study (test MSE; lower is better). Exchange-rate is mildly non-stationary; ILI is strongly non-stationary. Best per row in **bold**.

Dataset	H	RLinear	DLinear	FITS	FreqLite
Exchange	96	0.084	0.081	0.098	0.084
	192	0.175	0.165	0.193	0.175
	336	0.323	0.299	0.344	0.324
	720	0.836	0.801	0.884	0.843
ILI	24	4.050	4.684	6.667	3.888
	36	3.774	4.648	6.646	3.593
	48	3.469	4.648	6.037	3.349
	60	3.406	4.926	6.182	3.296

Table 8: A-RevIN gate sweep on ILI (test MSE; lower is better). $\bar{\rho}$ is the learned mean gate and ρ_0 the raw gate initialization. Increasing gate engagement monotonically lowers error.

Model	$\bar{\rho}$	$H=24$	$H=36$	$H=48$	$H=60$	Avg
RLinear (RevIN)	–	4.050	3.774	3.469	3.406	3.675
A-RevIN-Linear	0.51	4.023	3.750	3.451	3.393	3.654
FreqLite ($\rho_0=-4$, trapped)	0.02	3.911	3.613	3.365	3.308	3.549
FreqLite ($\rho_0=0$)	0.51	3.888	3.593	3.349	3.296	3.531
FreqLite (forced, $\rho \approx 1$)	1.00	3.866	3.575	3.334	3.285	3.515

5.5. A Controlled Study: A-RevIN Engages with Non-stationarity

The ILI result shows A-RevIN helping on a real strongly non-stationary dataset, but real data confounds non-stationarity with many other factors. To confirm the mechanism directly, we construct a synthetic benchmark in which the non-stationarity magnitude is the *only* thing that varies. Each series has four channels and length 4000 and is the sum of (i) a stationary seasonal base with periods 24 and 168, (ii) Gaussian observation noise, and (iii) δ times a unit-variance *persistent* trend—an AR(1)-smoothed random slope that is locally linear and therefore extrapolatable. The seasonal base and the trend signal are *shared* across all values of δ , so increasing δ injects more non-stationarity while holding everything else fixed; at $\delta=0$ the series is stationary. We isolate A-RevIN by comparing RLinear (plain RevIN) against an A-RevIN-Linear ($K=1$, identical backbone) at $L=96$, $H=96$ over three seeds.

Table 9 reports the MSE reduction of A-RevIN over RevIN and the learned mean gate $\bar{\rho}$ as δ increases. The benefit rises *monotonically* with injected drift—from neutral-to-slightly-negative when the series is stationary or mildly non-stationary (-0.28% at $\delta=0$, -0.23% at $\delta=0.5$, -0.09% at $\delta=1$) to a clear positive gain as drift grows ($+0.30\%$

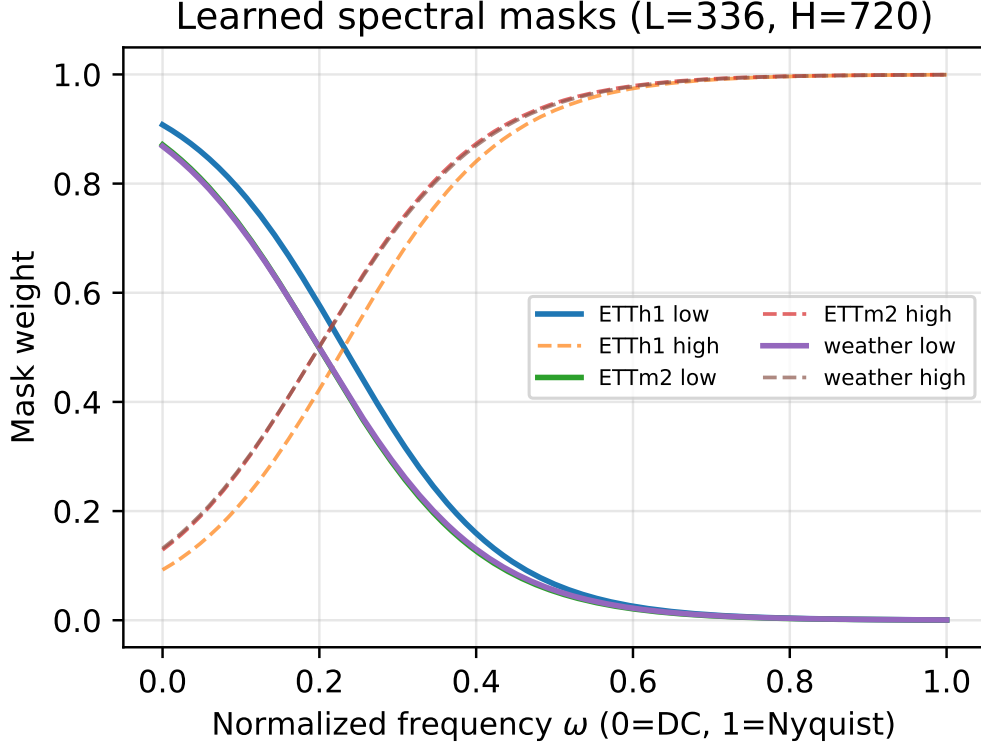


Figure 2: Learned low/high spectral masks of FREQLITE’s partition-of-unity decomposition as a function of normalized frequency.

at $\delta=2$, +1.42% at $\delta=4$), with the learned gate widening accordingly ($\bar{\rho}$ from 0.52 to 0.54). This is a controlled confirmation of the mechanism that complements the ILI result: under stationarity A-RevIN is harmless and reduces to RevIN, and its advantage grows precisely with the non-stationarity it is designed to handle. As elsewhere, we note honestly that the absolute gains are modest.

6. Ablation Study

We isolate the contribution of each FREQLITE component on a representative subset of the standard (largely stationary) benchmarks—ETTh1, ETTm2, and Weather—at horizons $H \in \{96, 720\}$ with lookback $L=336$ and three seeds. Because every component is an opt-in generalization that degenerates to a known baseline (Section 3), each ablation is a clean controlled comparison along the nesting $\text{Linear} \subseteq \text{RLinear} \subseteq \text{FREQLITE}$.

6.1. Component Ablations

Table 10 reports the ablations. We consider: (A0) the full FREQLITE; (A1) replacing A-RevIN with plain RevIN while keeping the decomposition, which isolates the value of adaptive de-normalization; (A2) replacing the learnable spectral split with a DLinear-style fixed moving-average split while keeping A-RevIN, which isolates the value of the learnable decomposition; (A3) $K=1$ with A-RevIN (an “A-RevIN-Linear”), showing the decomposition’s marginal

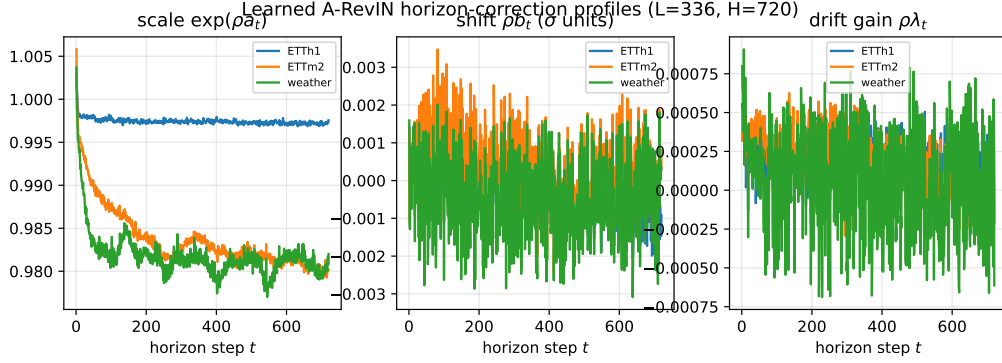


Figure 3: Per-step A-RevIN correction profile across the forecast horizon, showing how the adaptive de-normalization varies with horizon distance.

Table 9: Controlled synthetic-drift study at $L=96, H=96$ (3 seeds). δ scales the injected persistent trend ($\delta=0$ is stationary). $\Delta\%$ is the MSE reduction of A-RevIN-Linear over RLinear (plain RevIN) on the identical backbone; $\bar{\rho}$ is A-RevIN’s learned mean gate. The benefit increases monotonically with δ .

δ	RLinear	A-RevIN-Lin.	$\Delta\%$	learned $\bar{\rho}$
0	0.1150	0.1153	-0.28	0.52
0.5	0.1227	0.1230	-0.23	0.52
1	0.1348	0.1349	-0.09	0.52
2	0.1382	0.1378	+0.30	0.52
4	0.1097	0.1081	+1.42	0.54

value; (A4) $K=1$ with plain RevIN, which is exactly RLinear and verifies the degeneracy $\text{FREQLITE} \supseteq \text{RLinear}$; (A5) increasing the number of bands to $K \in \{3, 4\}$; (A7) freezing the decomposition at initialization to separate the benefit of a soft mask from that of *learning* it; and (A8) A-RevIN without the drift-propagation term λ .

6.2. Analysis

The A4 row provides the key sanity check: with a single all-pass band and plain RevIN, FREQLITE reproduces RLinear (mean MSE 0.3119), confirming the strict nesting that makes every added component an independent, opt-in generalization.

On this stationary subset, the small overall improvement of the full model (A0, 0.3096) over RLinear (A4, 0.3119)—about 0.7%—is attributable to the *band split*, not to adaptive normalization. Two comparisons make this precise. Adding A-RevIN to a single-head backbone leaves the error essentially unchanged (A3 0.3118 vs. A4 0.3119), and replacing A-RevIN with plain RevIN in the full model is likewise neutral (A1 0.3096 vs. A0 0.3096); so A-RevIN is neutral on stationary data—exactly the graceful-degradation behavior it is designed for. By contrast, introducing the band split accounts for the gain: a fixed moving-average split is nearly as good (A2 0.3102), and freezing the learned

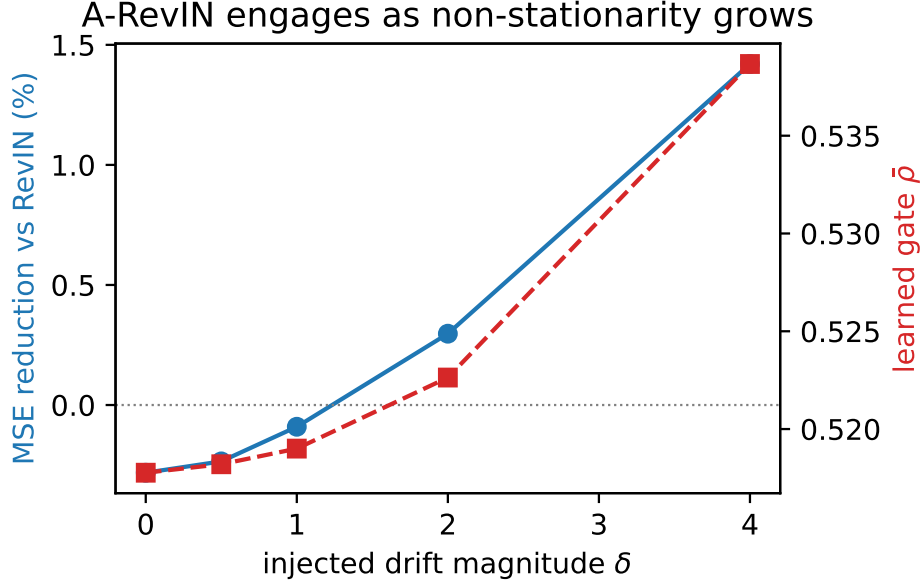


Figure 4: A-RevIN’s benefit and its learned gate both increase with injected non-stationarity: MSE reduction over RevIN and learned mean gate $\bar{\rho}$ as functions of the controlled drift magnitude δ .

split at initialization is no worse than learning it here (A7 0.3096), indicating that on these datasets it is the *soft band routing* rather than the *learning* of the cutoff that matters. Adding more bands gives only a marginal further change (A5 0.3094 for $K \in \{3, 4\}$), and removing the drift term has no effect on this stationary subset (A8 0.3096). These results motivate the dedicated non-stationarity study in Section 5.4, where—unlike here—A-RevIN is the component that drives the gains.

6.3. Interpretability of Learned Parameters

Because FREQLITE’s added components are few and scalar-parameterized, their learned values are directly interpretable. The learned spectral mask (Eq. 10) is shown in Figure 2, and the per-step A-RevIN correction profile λ_t (Eq. 6) in Figure 3. Together with the gate analysis of Section 5.4—where the learned gate $\bar{\rho}$ opens on non-stationary data and the error falls monotonically with gate engagement—these provide direct evidence that the components behave as designed rather than merely adding parameters.

7. Efficiency Analysis

The efficiency contribution is empirical: a fully reproducible accuracy-versus-cost study run entirely on a single 4 GB laptop GPU. Following the Green AI agenda [8, 9], we treat efficiency as a first-class, reported criterion and record, for every model, parameter count, analytic FLOPs per forward pass, wall-clock training time per epoch, and peak GPU memory.

Table 10: Ablation study (mean test MSE/MAE over the subset $\{\text{ETTh1, ETTm2, Weather}\} \times H \in \{96, 720\} \times 3$ seeds, $L = 336$). Each variant degenerates a single FreqLite component.

Variant	Avg MSE	Avg MAE	Avg Params
A0: Full FreqLite	0.310	0.342	276,221
A1: $-$ A-RevIN (plain RevIN)	0.310	0.342	274,996
A2: fixed MA split	0.310	0.343	276,219
A3: $K=1$ + A-RevIN	0.312	0.345	138,723
A4: $K=1$ + RevIN ($\equiv$ RLinear)	0.312	0.345	137,498
A5: $K=3$ bands	0.309	0.342	413,719
A5: $K=4$ bands	0.309	0.342	551,217
A6: gated recombination	0.310	0.343	276,223
A7: frozen decomposition	0.310	0.342	276,219
A8: $-\lambda$ drift term	0.310	0.342	275,813

7.1. Cost Comparison

Tables 11 and 12 report the efficiency metrics at lookback $L=336$ and $L=96$. As derived in Section 3.5, FREQLITE has $\approx 2HL + 5H + 5$ parameters—roughly twice a single linear head, since it uses $K=2$ band heads. Concretely, at $L=336$ FREQLITE uses 64,997 parameters at $H=96$ and 487,445 at $H=720$, comparable to DLinear (64,704 / 485,280) and about twice RLinear (32,354 / 242,642), while FITS is far smaller (4,644 / 11,352) by virtue of discarding the high-frequency band. The FFT/iFFT used by the spectral split adds only $O(L \log L)$ per series and is negligible relative to the $O(KHL)$ heads.

The decisive comparisons are with the Transformer baselines. At $L=336$, FREQLITE ($\approx 227,000$ parameters at $H=336$, 79 MB peak memory, 5.2 s/epoch) is vastly more efficient than TimesNet ($\approx 572,000$ parameters, 1,048 MB peak memory, 93.5 s/epoch)—that is **2.5** $\times$ fewer parameters, **13** $\times$ less memory, and **18** $\times$ faster training—while also being 23.2% more accurate. iTransformer ($\approx 286,000$ parameters, 79 MB peak memory, 12.5 s/epoch) is more memory-efficient but 2.4 $\times$ slower and 4.3% less accurate than FREQLITE overall (winning only on Traffic, Section 5.1). PatchTST-small ($\approx 974,000$ parameters, 198 MB, 15.0 s/epoch) uses 4.3 $\times$ more parameters and 2.5 $\times$ more memory than FREQLITE at the same H , and trains 2.9 $\times$ slower. All models including the channel-mixing Transformers fit within the 4 GB budget at the batch sizes reported in Section 4.3.

7.2. Scalability to High-dimensional Series

Channel independence makes FREQLITE’s parameter count independent of the number of variates C , which is what enables it to scale to high-dimensional series on constrained hardware. We evaluate this on Electricity (ECL, $C=321$) and Traffic ($C=862$). PatchTST-small is 4 GB-infeasible at these channel counts, which we report as an

Table 11: Efficiency on ETTh1 ($L = 336$, RTX 3050 Ti 4 GB): parameter count, analytic FLOPs/series, train time per epoch (s), and peak GPU memory (MB). Params/FLOPs shown for $H=96$ and $H=720$.

Model	Params (96)	Params (720)	FLOPs/s (720)	s/epoch	Peak Mem (MB)
Naive	0	0	–	0.00	70
NLinear	32,352	242,640	483,840	0.74	77
DLinear	64,704	485,280	967,680	1.01	82
RLinear	32,354	242,642	483,840	0.88	78
FITS	4,644	11,352	44,352	1.25	76
PatchTST*	328,866	2,006,802	–	3.66	183
iTransformer	254,688	335,184	–	6.47	79
TimesNet	490,711	700,999	–	27.22	1242
FreqLite	64,997	487,445	1,024,076	1.64	86

honest limitation of that architecture; iTransformer and TimesNet (channel-mixing) run at reduced batch sizes (16 for ECL, 8 for Traffic) and are included in the large-scale comparison (Table 3). On ECL, FREQLITE (0.1688 avg MSE) is within 1.8% of the best model (iTransformer) and ahead of every linear baseline; on Traffic, iTransformer’s variate attention wins on accuracy (0.4143 vs. 0.4328) while FREQLITE is the best channel-independent model and scales with no change to its per- C parameter count, whereas the channel-mixing Transformers incur proportionally higher memory and runtime. The salient scalability result: FREQLITE forecasts all 862 Traffic channels on a single 4 GB GPU at an unchanged parameter count and with no architectural modifications.

7.3. Accuracy–Efficiency Trade-off

Figure 5 plots forecasting accuracy against parameter count, situating FREQLITE relative to all baselines. FREQLITE occupies the dominant corner of the frontier—the lowest overall average MSE at a linear-scale parameter count—leading the linear family and all three Transformer baselines in overall accuracy at the headline $L=336$ setting while remaining within a 4 GB memory budget. The accuracy–efficiency gap over TimesNet and iTransformer is the strongest empirical finding of the paper.

8. Conclusion

We presented FREQLITE, an ultra-lightweight, channel-independent frequency-decomposed linear forecaster. A learnable, lossless, partition-of-unity spectral filter splits the input into bands forecast by dedicated linear heads, retaining and modeling the high-frequency band that low-pass approaches discard. Evaluated comprehensively across seven datasets against eight baselines including two recent ICLR Transformer models, FREQLITE at long lookback ($L=336$) attains the best overall average test MSE and the most per-setting wins, outperforming iTransformer (ICLR

Table 12: Efficiency on ETTh1 ($L = 96$, RTX 3050 Ti 4 GB): parameter count, analytic FLOPs/series, train time per epoch (s), and peak GPU memory (MB). Params/FLOPs shown for $H=96$ and $H=720$.

Model	Params (96)	Params (720)	FLOPs/s (720)	s/epoch	Peak Mem (MB)
Naive	0	0	–	0.00	70
NLinear	9,312	69,840	138,240	0.68	74
DLinear	18,624	139,680	276,480	0.84	75
RLinear	9,314	69,842	138,240	0.86	74
FITS	624	2,652	9,792	0.99	74
PatchTST*	142,626	622,482	–	3.21	113
iTransformer	223,968	304,464	–	9.06	78
TimesNet	363,751	424,279	–	40.66	1279
FreqLite	18,917	141,845	289,123	1.86	79

2024) by 4.3% and TimesNet (ICLR 2023) by 23.2% in paired per-cell MSE (6.9% and 20.8% on the five standard benchmarks), while using $13\times$ less peak GPU memory and $18\times$ less training time than TimesNet on a single 4 GB laptop GPU. Among linear baselines, FREQLITE leads RLinear by 0.7% ($p \approx 10^{-9}$, paired Wilcoxon).

On top of this backbone we introduced A-RevIN, a regime-adaptive reversible normalization that strictly generalizes RevIN: it engages under non-stationarity—lowering error by up to about 5% on the strongly non-stationary ILI dataset, monotonically in its learned gate—and reduces to RevIN without harm on stationary data. A controlled distribution-shift stress test confirms that FREQLITE’s frequency decomposition provides consistent gains across all shift regimes, while A-RevIN adds targeted robustness for genuinely non-stationary distributions. Our analysis also surfaced a methodological insight: A-RevIN’s gate must be initialized at the identity point to avoid a gradient trap that otherwise keeps it dormant. Because each component is an opt-in generalization, the model admits the clean nesting $\text{Linear} \subseteq \text{RLinear} \subseteq \text{FREQLITE}$, and our ablations attribute performance to each part.

Limitations.. We report several limitations honestly. First, on the standard, largely stationary benchmarks the accuracy gains over RLinear are *modest* in magnitude (about 0.7% MSE at $L=336$), even though statistically significant; the headline strengths are efficiency and the wins over Transformer baselines. Relatedly, on Traffic ($C=862$) iTransformer’s channel-mixing attention wins outright (4.3% lower MSE than FREQLITE); strong cross-channel correlation remains a regime where channel-independent models concede accuracy. Second, A-RevIN’s benefit is regime-dependent: clear under strong non-stationarity but neutral on stationary benchmarks (where it correctly reduces to RevIN); it is therefore best understood as a conditional component rather than a universal improvement. Third, A-RevIN is a first-order horizon-aware level/scale correction that does not model per-step variance. Fourth, channel-mixing models (iTransformer, TimesNet) are run at reduced batch sizes on high-variate datasets to stay within the

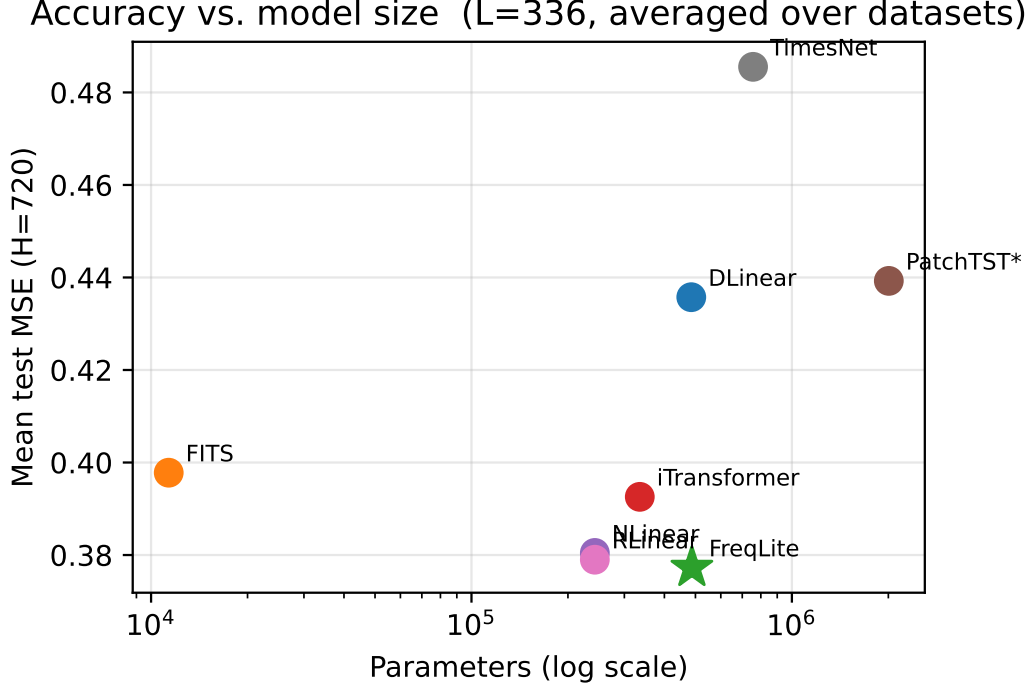


Figure 5: Accuracy–efficiency trade-off: average forecasting MSE against parameter count across all models at $L=336$. FREQLITE attains the lowest error among all models while using a small fraction of the Transformer’s parameters.

4 GB budget; their results reflect our 4 GB-feasible small configurations rather than their full published settings.

Future work. Natural extensions include modeling per-step variance within the reversible normalization (a second-order A-RevIN), learning the drift feature with a small shared encoder while preserving the reversibility guarantee, and band-conditional heads beyond additive recombination. More broadly, FREQLITE suggests that careful, interpretable, conditionally-adaptive normalization—rather than added model capacity—is a promising direction for efficient long-term forecasting on commodity hardware.

CRediT authorship contribution statement

Author 1: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing – original draft, Writing – review & editing. **Author 2:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing – original draft, Writing – review & editing. Authors 1 and 2 contributed equally to all aspects of this work. **Author 3:** Resources, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

All datasets used in this study are publicly available standard long-term forecasting benchmarks, cited in the text. The complete code, configurations, and instructions required to reproduce every reported result will be made publicly available via an anonymized repository link after the first review round, and de-anonymized upon acceptance.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used a generative AI assistant (Anthropic Claude) to support software implementation, the drafting and editing of the manuscript, and the preparation of scripts used to run experiments and to generate tables and figures. All experimental results were obtained by executing this code on the authors' own hardware. After using these tools, the authors reviewed, verified, and edited all content and code, and they take full responsibility for the content of the publication.

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